MATH INQUIRERS HOLIDAY PROGRAM

Free 30-minute taster

Lesson 1: Code the Pattern (Sequences and the nth Term) • Math Inquirers Level 6 (ages 11-13) • Maths + coding bridge

Notice it. Test it. Explain why.

What this is

A short sample of a real Math Inquirers lesson for Level 6 (ages 11-13, after the PSLE). It is Lesson 1 of the Level 6 holiday program (Code the Pattern), shortened to about 30 minutes: a warm-up, a hands-on explore, a mini-lesson, a few worksheet questions and an exit ticket. The same routine runs through all ten 90-minute lessons.

The 30 minutes

Time	Part	What happens
5 min	**Warm-up**	Next! Pattern quick-fire: six sequence cards (add, take away, square numbers, doubling, Fibonacci, decimals). Pairs type the next two terms in the chat and say the rule if they spot it. No racing and no points.
8 min	**Explore**	Pairs build matchstick triangles in a row (3, 5, 7 sticks), make a table, predict the sticks for pattern 10 and pattern 100 without building them, then check pattern 10 with a loop and a variable in Scratch.
5 min	**Mini-lesson**	Term-to-term rule (add 3) versus the rule for the nth term (3n + 2): the difference tells you the times table, then compare to find what to add or take away. Find the 100th term. Test it in code: set term to the 1st term, repeat 99, change term by 3, say term. The 100th term is only 99 jumps from the 1st. Before page 1: work backwards to decide whether a number is in a sequence. Before page 2: trace a loop in a table, plan a program (say first, then change) and find a jump-count bug.
9 min	**Worksheet**	The child works on questions 1 to 4 of the Lesson 1 sheet (next pages), then checks them with the tutor.
3 min	**Exit ticket**	The sequence 8, 11, 14, 17, ... (a) Write the rule for the nth term. (b) Find the 100th term. (c) A loop starts at the 1st term. How many times must it repeat "change by 3" to reach the 100th term?

What you need

Laptop or desktop with Scratch 3 open (scratch.mit.edu, no account needed), about 25 matchsticks or toothpicks per child (or squared paper), mini whiteboard, workbook. Optional: a Python editor for children who already know Python.

Online: The tutor shares the slides and a Scratch window. Share your Scratch window and build the four blocks together on screen; the blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Scratch is free at scratch.mit.edu and needs no account in class. Children hold paper up to the camera or share their screen when asked. Matchsticks can be drawn as lines.

Pages in this file: this page, the child's worksheet (about two pages), then tutor notes with the answers. Do not give the tutor-notes pages to the child. This taster is enrichment for thinking skills; it does not promise exam results and is not affiliated with MOE or the Gifted Education Programme.

CODE THE PATTERN

Name _____ Group _____ Date _____

No.1

Matchstick squares. Each new pattern adds one more square. Fill in the table.



Pattern 1 Pattern 2 Pattern 3

Pattern n	1	2	3	4	5	6
Sticks	4					

a Each new square adds _____ sticks. Why not 4? _____

b Complete the rule: sticks = _____ × n + _____

c Pattern 100 has _____ sticks. (d) Which pattern has 61 sticks? _____

No.2

For each sequence, write the difference, the rule for the nth term and the 100th term. (Two more are on the Think harder page.)

Sequence	Difference	nth term	100th term
(a) 6, 10, 14, 18, ...			
(b) 1, 6, 11, 16, ...			

No.3

Look at the sequence 5, 9, 13, 17, ...

a nth term: _____ (b) Is 101 a term? Which one? _____

c Is 150 in the sequence? Explain. _____

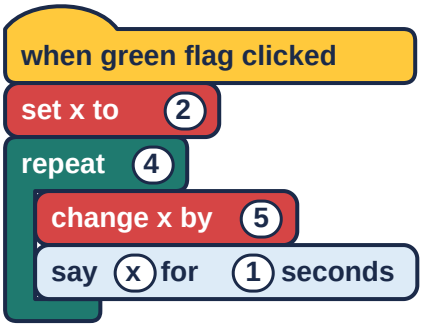
```
term = 5
for pos in range(1, 101):
    print(pos, term)
    term = term + 3
```

Python corner (optional): a term tester

It prints the first 100 terms of 5, 8, 11, ... Change the 5 and the 3 to check your answers to Q1 and Q2.

No.4

Predict, then trace. What does this code make the sprite say? Fill in the trace table.



Loop	x after "change"
1	
2	
3	
4	

(a) The last number it says: _____

b With repeat 100, what is the last number? Explain without running it.

Tutor notes: Lesson 1, Code the Pattern

Sequences and the nth Term • Level 6 (ages 11-13, after the PSLE)

Goal

Continue a sequence, find the rule for the nth term of a sequence with a constant difference, use it to find far terms such as the 100th, work backwards to decide whether a number is in a sequence, and test predictions with a loop and a variable in Scratch.

Skills: Pattern spotting, generalising with a letter, predicting, testing in code, debugging (counting the jumps)

Opening

Welcome to Math Inquirers Level 6! The PSLE is done, so there is no exam today: just puzzles, patterns and code.

Every lesson we notice a pattern, test it in code, and explain why. Today's question: how can you find the 100th number in a pattern without writing out all 100?

Warm-up (about 5 minutes)

Next! Pattern quick-fire: six sequence cards (add, take away, square numbers, doubling, Fibonacci, decimals). Pairs type the next two terms in the chat and say the rule if they spot it. No racing and no points.

Pairs. I will show a sequence card. Take your time and type the next two terms in the chat. If you spot the rule, say it. There is no race and there are no points.

Careful: not every card adds the same amount each time.

Explore (about 8 minutes)

Pairs build matchstick triangles in a row (3, 5, 7 sticks), make a table, predict the sticks for pattern 10 and pattern 100 without building them, then check pattern 10 with a loop and a variable in Scratch.

Predict: Pattern 1 is a triangle made of 3 matchsticks. Each new pattern adds one more triangle in the row. Before you build anything: how many sticks will pattern 100 need? Write your guess in the corner of your whiteboard.

- Stuck building: lay the triangles point up, point down, point up, so each new triangle shares a stick.
- Stuck on pattern 100: "How many sticks does each new triangle add? How many new triangles come after the first one?"
- Finished early: in Scratch, set sticks to 3, repeat 9, change sticks by 2, say sticks. Does it match your pattern 10? Now change it to reach pattern 100.

Mini-lesson (about 5 minutes)

Term-to-term rule (add 3) versus the rule for the nth term (3n + 2): the difference tells you the times table, then compare to find what to add or take away. Find the 100th term. Test it in code: set term to the 1st term, repeat 99, change term by 3, say term. The 100th term is only 99 jumps from the 1st. Before page 1:

work backwards to decide whether a number is in a sequence. Before page 2: trace a loop in a table, plan a program (say first, then change) and find a jump-count bug.

Worksheet answers (questions 1 to 4)

Q1	(a) 4, 7, 10, 13, 16, 19. Each new square adds 3 sticks, not 4, because it shares one side with the square before it. (b) sticks = 3 × n + 1 (c) 301 (d) Pattern 20 (3 × 20 + 1 = 61)
Q2	(a) difference 4; 4n + 2; 402 (b) 5; 5n − 4; 496 (the other two sequences of this question are now Think harder 4)
Q3	(a) 4n + 1 (b) Yes. 101 − 1 = 100 = 4 × 25, so 101 is the 25th term. (c) No. Every term is 1 more than a multiple of 4, so every term is odd; 150 is even. (150 − 1 = 149 is not a multiple of 4.)
Q4	Trace: x is 7, 12, 17, 22. (a) 22 (b) 502. x starts at 2 and 100 jumps of 5 add 500.

Talk time

Ali says the 100th term of 7, 10, 13, ... is 7 + 3 × 100 = 307. Is he right? Explain with a rule, a picture or code.

Answer: No. The 1st term is already 7 before any jump, so the 100th term is only 99 jumps away: 7 + 3 × 99 = 304. The rule 3n + 4 gives 304 too. The rule is fastest for one far term; code is best for listing many terms or checking an answer.

Exit ticket (about 3 minutes)

The sequence 8, 11, 14, 17, ... (a) Write the rule for the nth term. (b) Find the 100th term. (c) A loop starts at the 1st term. How many times must it repeat "change by 3" to reach the 100th term?

Answer: (a) 3n + 5 (b) 305 (c) 99 times

Watch for these mistakes

- Using only the difference as the rule (writing 3n for 5, 8, 11, ...) and not comparing with the 1st term.
- Off by one: doing 100 jumps from the 1st term (5 + 3 × 100) instead of 99, in the maths and in the repeat block.
- Putting "change" before "say" in a loop, so the sprite never says the 1st term.

Extension for fast finishers

Make the program say the position and the term together (for example "term 7 is 23") using a second variable pos and the join block. Then make it stop at the first term bigger than 1000.

After the taster

Tell the parent one thing their child noticed and one thing they explained well. Then share the next steps: [dates, fee and sign-up link]. The free diagnostic for the matching level helps choose between Level 6 (ages 11-13, after the PSLE) and the level either side (see the placement note).